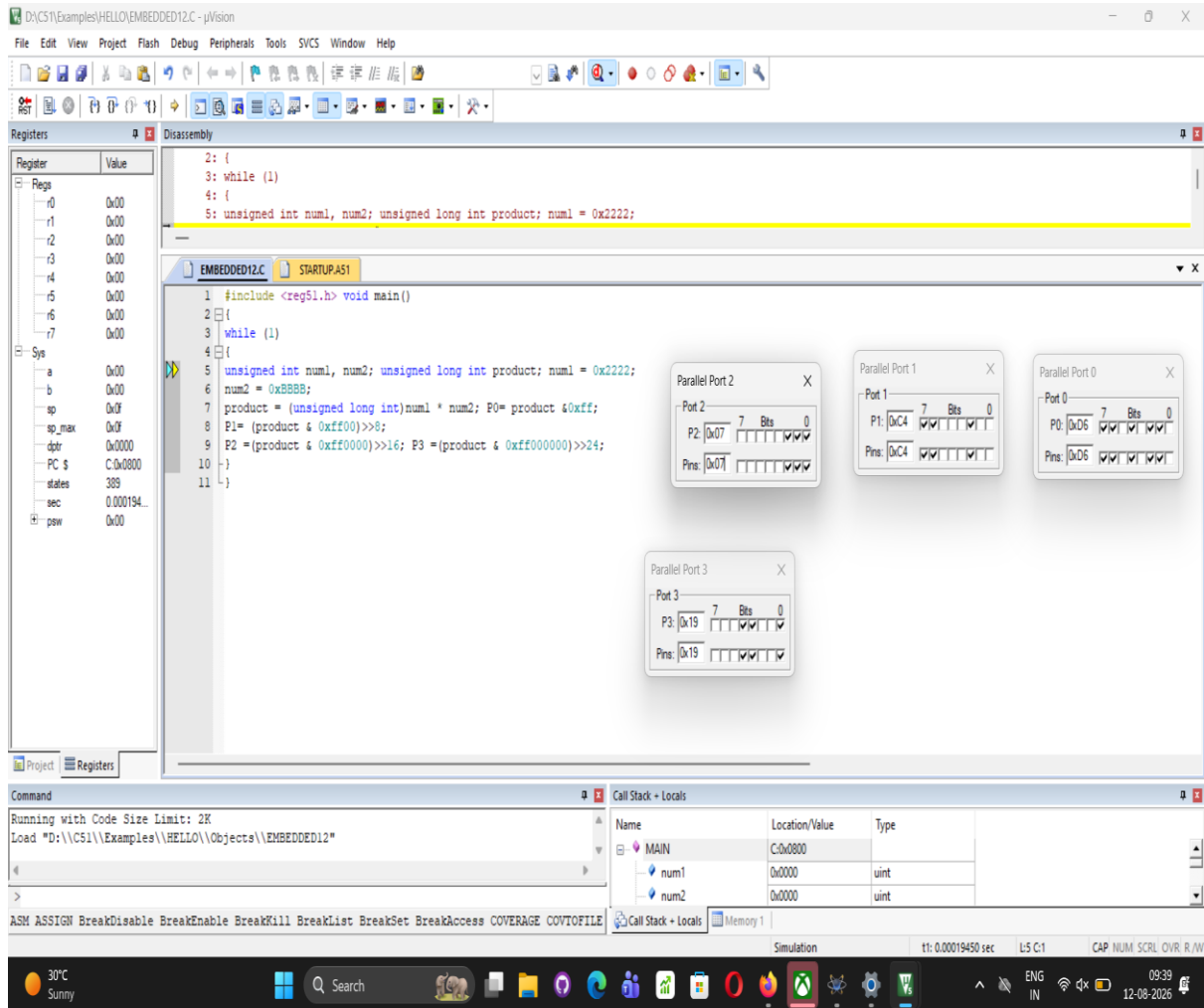


Exp No: 12

EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS



The screenshot displays the Keil uVision IDE interface for an Embedded C program. The main window shows the source code for a program that multiplies two 16-bit binary numbers. The code is as follows:

```
1 #include <reg51.h> void main()
2 {
3     while (1)
4     {
5         unsigned int num1, num2; unsigned long int product; num1 = 0x2222;
6         num2 = 0xB8BB;
7         product = (unsigned long int)num1 * num2; P0= product & 0xff;
8         P1= (product & 0xff00)>>8;
9         P2 =(product & 0xff0000)>>16; P3 =(product & 0xff000000)>>24;
10    }
11 }
```

The Registers window on the left shows the current state of the registers. The Command window at the bottom shows the simulation progress, including the time taken (0.00019450 sec) and the location of the program (C:\0800).

The Call Stack window shows the current function being executed, which is the MAIN function. The variables num1 and num2 are listed with their values (0x0000 and 0x0000 respectively).

The Parallel Port windows (Parallel Port 0, Parallel Port 1, Parallel Port 2, and Parallel Port 3) show the current values of the ports. The values are as follows:

- Parallel Port 0: P0: 0x06, Pins: 0x06
- Parallel Port 1: P1: 0x04, Pins: 0x04
- Parallel Port 2: P2: 0x07, Pins: 0x07
- Parallel Port 3: P3: 0x19, Pins: 0x19